

# KARTHIK VAKKALAGADDA

Software Engineer in Test (SDET) | Test Automation & Test Infrastructure

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## SUMMARY

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- **7+ years** building test automation frameworks and CI/CD test infrastructure for high-volume digital ordering, POS, and payments platforms across four Inspire Brands properties — Web, Mobile, and API — orchestrated through GitLab CI with BrowserStack-backed parallel execution.
- Took framework ownership early: authored the majority of the Java automation framework for Arby's and Buffalo Wild Wings, growing those suites to 2,000+ automated tests, and owned Web and API automation single-handedly within 18 months of entering the field — now technical lead for a 4-6 person distributed US/India team.
- Sole architect and developer of a unified QA platform (personal project) built on MCP servers, agent skills, and prompt-driven test generation, with working GitLab CI and Jira/Xray integrations; sanitized rebuild published publicly on GitHub (shuriken, play-left).

## CORE COMPETENCIES

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- Test infrastructure & CI/CD pipeline design • test orchestration • quality gates • parallel and cross-device execution
- Automation framework architecture across Web, Mobile, and API • shared runners • reusable flow libraries
- Resiliency and fault-tolerance testing (circuit breakers, induced-failure scenarios)
- Observability-driven root-cause analysis using New Relic and Splunk — log correlation, custom metrics/dashboards, deployed-version verification
- AI-assisted engineering tooling: MCP servers, agent skills and tool-use workflows, prompt-driven test generation, LLM-based evaluation
- POS integration testing across third-party platforms (Micros, Infor, NCR, BrinkPOS) • in-store and digital ordering flows
- Payments and transaction validation: payment and pricing flows, order totals, and POS-to-platform reconciliation
- A/B test and feature-flag validation (Optimizely, Contentful)
- Technical leadership, merge-request review at volume, and mentoring across onshore/offshore teams

## TECHNICAL TOOLS

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**Languages:** Java, Python, JavaScript (via Playwright), SQL

**Automation & Frameworks:** Selenium WebDriver, Appium, Playwright, Maestro (incl. Maestro AOS TE), REST Assured, TestNG, Cucumber BDD (Gherkin), Tosca

**Test Infrastructure & CI/CD:** GitLab CI/CD, Docker, Gradle, Git, BrowserStack device cloud, secrets management, test orchestration, parallel execution

**Observability & Experimentation:** New Relic, Splunk (log-based root-cause analysis, custom metrics, dashboards), Optimizely, Contentful (A/B testing, feature-flag validation)

**AI-Assisted Tooling:** MCP servers, agent skills, Claude, GitHub Copilot, prompt-driven test generation, LLM-based evaluation frameworks

**Databases & APIs:** SQL, MongoDB, REST APIs, Postman, Swagger

**Cloud/Messaging (integration & validation):** Azure Service Bus, Azure-hosted databases

**Test & Delivery Management:** Jira, Xray, qTest, Confluence, ReportPortal

**POS & Payments:** POS integration testing (Micros, Infor, NCR, BrinkPOS), payments and pricing validation, POS-to-platform reconciliation, system integration testing

**Specialized Testing:** Resiliency & fault-tolerance testing, accessibility testing (UsableNet, VoiceOver, TalkBack), UAT

**Other:** Figma

## PROFESSIONAL EXPERIENCE

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### SDET / Automation Technical Lead — Multi-Brand Digital Ordering & POS Platforms

Capgemini (Client: Inspire Brands) | Apr 2026 – Present | Atlanta, GA

- Expanded from a single-brand to a four-brand remit — Sonic, Buffalo Wild Wings, and Dunkin', plus selected Arby's work — owning automation standards, framework direction, and CI/CD test execution across their digital ordering and POS platforms.
- Built Sonic E2E regression coverage using AI agent skills and prompt-driven test generation, establishing a standardized workflow for authoring and reviewing new automated tests across the team.
- Dunkin' loyalty platform: resiliency and fault-tolerance testing, including circuit-breaker validation under induced failure conditions, using Splunk for log-based root-cause analysis on service and third-party integration failures.
- Buffalo Wild Wings: automation coverage for the Business Service Layer (BSL) change, validating API contracts, regression, and integration paths across the service-layer migration.
- Technical lead for a 4–6 person US-onsite / India-offshore automation team — defining framework standards, test infrastructure, and reporting conventions, mentoring engineers, and driving adoption of shared tooling.

### SDET / Automation Lead — Sonic E2E Digital Ordering & POS Platform

Capgemini (Client: Inspire Brands) | Jan 2024 – Mar 2026 | Atlanta, GA

- Owned test strategy and automation architecture for Sonic's end-to-end digital ordering and POS integration platform across Web, Mobile, and API layers.
- Architected the end-to-end CI/CD test infrastructure: GitLab pipelines integrating BrowserStack execution, quality gates, and Slack/email reporting for parallel execution and fast feedback, improving release visibility and removing manual release prep.
- Designed and built the Maestro mobile E2E framework, defining shared flows, smoke suites, and critical-path coverage; introduced standardized runners (Playwright for web, Maestro AOS TE for Android) reused across brands and platforms.
- Used Splunk for log-based root-cause analysis on service and third-party integration failures, supporting circuit-breaker and error-scenario testing with custom metrics, dashboards, and deployed-service-version verification.
- Validated A/B tests and feature-flag rollouts using Optimizely and Contentful as part of the Sonic digital ordering experience.
- Encountered Azure Service Bus and Azure-hosted databases during integration testing and data validation.
- Drove repository and domain restructuring — config registry, loader updates, security hardening — and standardized deployment flows across automation and menu domains.
- Performed accessibility testing with UsableNet and native iOS/Android tooling (VoiceOver, TalkBack) to validate compliance across mobile and web surfaces.

### SDET — Sonic POS, Payments & Digital Integration

Capgemini (Client: Inspire Brands) | May 2023 – Dec 2023 | Atlanta, GA

- Built and maintained automated regression and smoke suites across Web, Mobile, and API layers, validating integrations between digital ordering platforms and POS systems (Micros, Infor) across ordering, payment, and pricing workflows.
- Backend API testing with REST Assured and data-integrity validation in SQL and MongoDB; reconciled POS records against ordering-platform and processor data to catch pricing and payment mismatches.
- Partnered with stakeholders through UAT to define test requirements and validate in-store operational workflows.

### SDET — Buffalo Wild Wings (BWW Choice)

Capgemini (Client: Inspire Brands) | May 2022 – Apr 2023 | Bengaluru, India

- Sole automation engineer for BWW Choice mobile web and API layers — built and maintained the in-store dining automation suites in Java with Selenium and REST Assured, integrated into GitLab CI/CD with ReportPortal and Jira reporting.
- Used New Relic for log-based root-cause analysis on service integration failures during regression cycles.
- Extended the shared automation framework as it moved across brands, keeping runners and reporting conventions consistent.

## Automation Engineer, Web & API Lead — Arby's & Buffalo Wild Wings

Capgemini (Client: Inspire Brands) | Jul 2020 – Apr 2022 | Bengaluru, India

- Authored the majority of the Java automation framework (Selenium, Appium, REST Assured, Cucumber BDD, TestNG) from its early stages — flow structure, reusable utilities, test data handling, and reporting — as a first-year engineer.
- Owned Web and API automation end to end as the sole engineer on both layers and co-owned mobile app automation with one other engineer, growing the Arby's and Buffalo Wild Wings suites to 2,000+ automated tests.
- Used New Relic for log-based root-cause analysis on service and third-party integration failures affecting regression suites.
- Reviewed and merged 200+ merge requests, resolving conflicts across a shared multi-brand codebase, and onboarded new joiners onto the framework.
- Configured GitLab CI/CD pipelines for scheduled and on-demand execution with multiple configurable run setups (brand, environment, and suite selection), giving the team repeatable runs without manual setup.

## Test Automation Engineer — T-Mobile Customer Care & Retail

Capgemini (Client: T-Mobile) | Jan 2020 – Jun 2020 | Bengaluru, India

- Tested desktop, web, mobile, and API applications for retail and care operations using Tosca and Appium (Java); validated APIs with Postman and tracked defects in Jira.

## QA Automation Engineer

Capgemini | Aug 2019 – Dec 2019 | Pune, India

- Tested web and Windows desktop applications using UFT (Unified Functional Testing), building foundational test-automation skills before moving to API and mobile testing on the T-Mobile account.

## ENGINEERING PROJECTS & PUBLIC PORTFOLIO

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- **Unified QA Platform (personal project)** — Independently designed and built the platform architecture as sole developer in personal time — test orchestration, CI/CD tooling, Jira/Xray integration, and a web dashboard, plus supporting utilities (Gherkin-to-Jira importer, health checks, Xray coverage dashboards) across Sonic, Arby's, and Buffalo Wild Wings.
- **AI-Assisted Test Generation** — Built prompt-driven test generation runners, MCP servers, agent skill libraries, and EPIC/TE testcase creation runners on top of the platform, redesigning the test-authoring workflow end to end.
- **shuriken (public, GitHub)** — Sanitized public rebuild of the platform's core automation architecture with dummy data, including its MCP servers, agent skills, and prompt-driven runners.
- **play-left (public, GitHub)** — Standalone multi-brand Playwright web E2E framework, split out of shuriken as a companion repository.
- **Personal Tracker (public, GitHub)** — TypeScript/Vite app with local-first IndexedDB storage, tested with Playwright/Vitest and deployed via GitHub Actions; live demo linked from portfolio.
- **Cricket Field Planner (public, GitHub)** — Kotlin/Android app paired with a React/TypeScript companion, using the Gemini API and deployed on Cloudflare Pages; live demo linked from portfolio.

## CERTIFICATIONS & EDUCATION

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**Certifications:** AWS Certified Cloud Practitioner • AWS Generative AI Essentials • Tricentis Tosca L1 & qTest • Harvard ManageMentor

**Education:** B.Tech, Electronics & Communication Engineering — IIITDM Jabalpur, India, 2019